

ADAS Assignment

Experiment 1

AEL ZG534

Name and ID of the Student : Neelam Singh (2024HT65640)



Work Integrated Learning Programme
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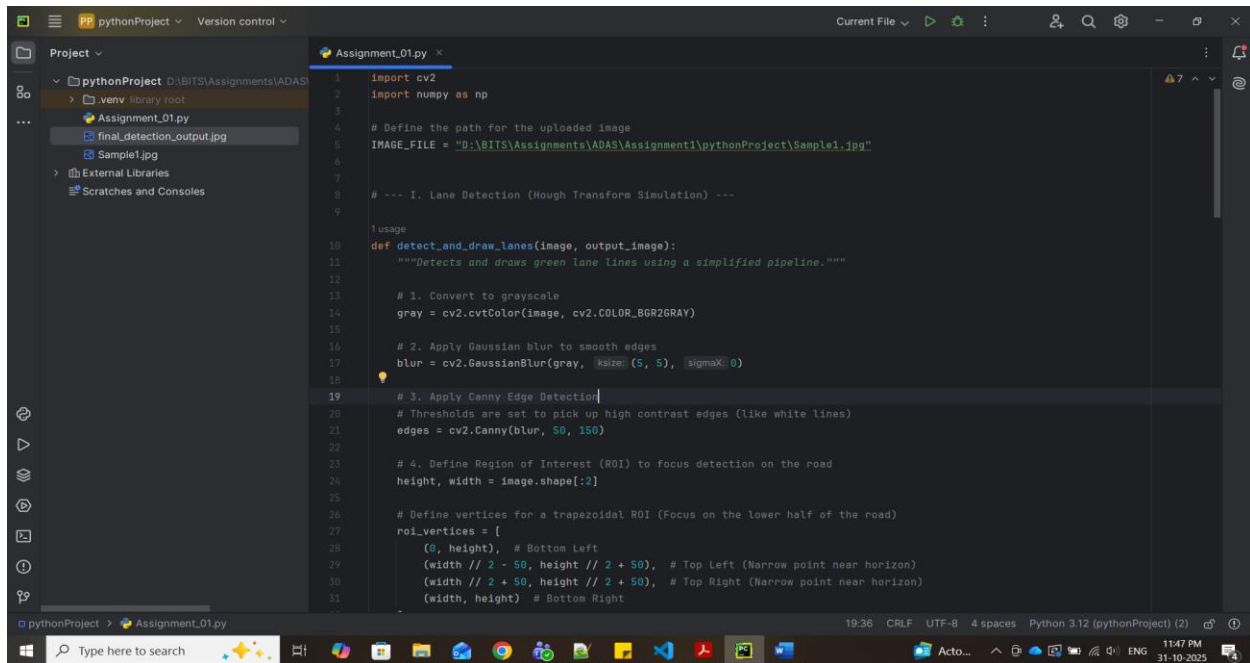
Python Script with OpenCV library to perform below tasks –

I. Draw Red color Rectangle Boxes on Red Suv and White Car

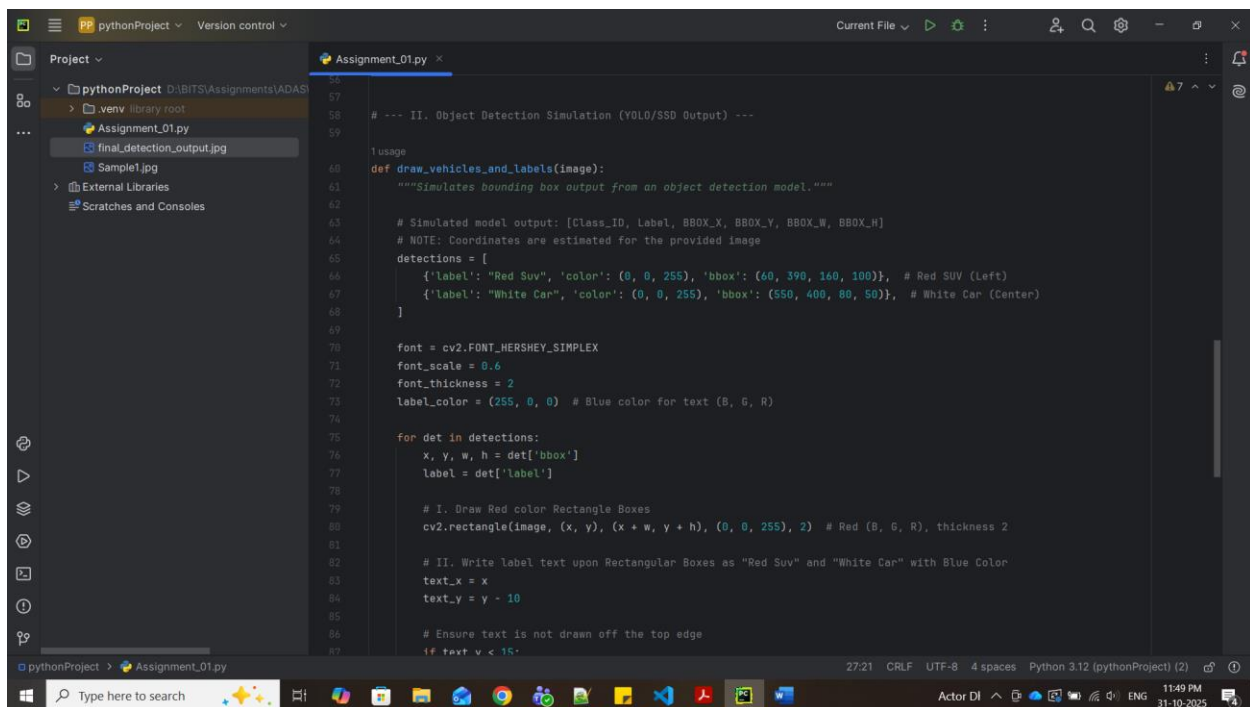
II. Write label text upon Rectangular Boxes as “Red Suv” and “White Car” with Blue Color

III. Draw the straight Lane lines with Green color

Code Snippet:



```
1 import cv2
2 import numpy as np
3
4 # Define the path for the uploaded image
5 IMAGE_FILE = "D:\\BITS\\Assignments\\ADAS\\Assignment1\\pythonProject\\Sample1.jpg"
6
7 # --- I. Lane Detection (Hough Transform Simulation) ---
8
9 usage
10 def detect_and_draw_lanes(image, output_image):
11     """Detects and draws green lane lines using a simplified pipeline."""
12
13     # 1. Convert to grayscale
14     gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
15
16     # 2. Apply Gaussian blur to smooth edges
17     blur = cv2.GaussianBlur(gray, ksize=(5, 5), sigmaX=0)
18
19     # 3. Apply Canny Edge Detection
20     # Thresholds are set to pick up high contrast edges (like white lines)
21     edges = cv2.Canny(blur, 50, 150)
22
23     # 4. Define Region of Interest (ROI) to focus detection on the road
24     height, width = image.shape[:2]
25
26     # Define vertices for a trapezoidal ROI (Focus on the lower half of the road)
27     roi_vertices = [
28         (0, height), # Bottom Left
29         (width // 2 - 50, height // 2 + 50), # Top Left (Narrow point near horizon)
30         (width // 2 + 50, height // 2 + 50), # Top Right (Narrow point near horizon)
31         (width, height) # Bottom Right
```



```
55
56
57 # --- II. Object Detection Simulation (YOLO/SSD Output) ---
58
59 usage
60 def draw_vehicles_and_labels(image):
61     """Simulates bounding box output from an object detection model."""
62
63     # Simulated model output: [Class_ID, Label, BBOX_X, BBOX_Y, BBOX_W, BBOX_H]
64     # NOTE: Coordinates are estimated for the provided image
65     detections = [
66         {'label': "Red Suv", 'color': (0, 0, 255), 'bbox': (60, 390, 160, 100)}, # Red SUV (Left)
67         {'label': "White Car", 'color': (0, 0, 255), 'bbox': (550, 400, 80, 50)}, # White Car (Center)
68     ]
69
70     font = cv2.FONT_HERSHEY_SIMPLEX
71     font_scale = 0.6
72     font_thickness = 2
73     label_color = (255, 0, 0) # Blue color for text (B, G, R)
74
75     for det in detections:
76         x, y, w, h = det['bbox']
77         label = det['label']
78
79         # I. Draw Red color Rectangle Boxes
80         cv2.rectangle(image, (x, y), (x + w, y + h), (0, 0, 255), 2) # Red (B, G, R), thickness 2
81
82         # II. Write label text upon Rectangular Boxes as "Red Suv" and "White Car" with Blue Color
83         text_x = x
84         text_y = y - 10
85
86         # Ensure text is not drawn off the top edge
87         if text_y < 15:
```

Output:

